Name_____ Date_____ Period_____

LESSON 2-6 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Median and Outliers

Maryland Standard 6.SP.B.5d · Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.

REMEMBER

- An **outlier** is a value much greater or much less than the rest of the data.
- An outlier pulls the **mean** toward it. The **median** barely moves, so it is **resistant** to outliers.
- With an outlier, the median usually describes a **typical value** better. Still report the outlier; it is real data.

WORKED EXAMPLE

Six bike rides lasted 14, 18, 15, 16, 64, and 17 minutes. The mean is 24 minutes. Which measure better describes a typical ride?

Step 1 Order: 14, 15, 16, 17, 18, 64. The outlier is 64.

Step 2 Median: (16 + 17) ÷ 2 = 16.5. The mean, 24, is greater than 5 of the 6 rides.

Answer **The median, 16.5 minutes**

WORD BANK

outlier – a number far away from the others
median – the middle number
mean – add all, then divide by how many
typical – usual, like most of the data

PRACTICE

show your work · label units

1. Find the median of each data set. Circle the outlier in each set.

Step 1 Order each list.

Step 2 Circle the number far from the rest.

Step 3 Find the middle number.

a. 9, 11, 10, 12, 13, 8, 48 _____

b. 3, 70, 72, 75, 78, 71, 74 _____

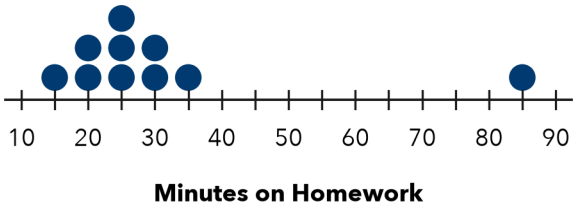
c. 25, 28, 30, 31, 33, 95 _____

d. 40, 41, 43, 44, 46, 47, 150 _____

e. 6, 7, 7, 8, 9, 10, 34 _____

f. 56, 52, 55, 54, 9, 53, 57, 58 _____

2. Homework Minutes. The dot plot shows minutes 10 students spent on homework one night. The mean is 31 minutes.



Minutes on Homework

Step 1 Find the dot far from the others.

Step 2 Order the 10 values; the median is between the 5th and 6th.

Step 3 Count students below 31.

a. Outlier: _____

b. Median: _____ min

c. Students below the mean: _____

d. Better center: _____

3. Library Visitors. Visitors on 7 days: 64, 58, 61, 66, 59, 63, and 4 (the library closed early for snow).

a. Median of all 7 days: _____ **b.** Median without the 4: _____

4. Car Wash. A club earned $10, $12, $13, $14, $15, $16, and $60 on 7 Saturdays.

Step 1 Mean: add the 7 amounts, then divide by 7.

Step 2 Median: the 4th amount in order.

a. Mean: $ ____ **b.** Median: $ ____ **c.** Better center: _____

5. Find the Error. Andre's data are 31, 33, 34, 36, 37, 39, and 210. He says, "The outlier 210 makes the median much bigger, so the median is not a good measure." Is he right? Find the median to support your answer.

Frame *Andre is ___. The median is ___. Without 210 it would be ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home about a normal trip and their slowest trip. Which trip is an outlier?

Name_____ Date_____ Period_____

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Comparing Distributions

Maryland Standard 6.SP.B.5d · Learning target – I can compare two distributions using the median and the interquartile range together.

REMEMBER

- Compare **centers** with the medians. Compare **spreads** with the IQRs (IQR = Q3 − Q1).
- A smaller IQR means more **consistent** data. A full comparison names **both** measures for **both** groups, with numbers.

WORKED EXAMPLE

Team Red: median 14 points, IQR 6. Team Blue: median 18 points, IQR 15. Compare the teams.

Step 1 Centers: 18 > 14, so Blue usually scores more.

Step 2 Spreads: 6 < 15, so Red is more consistent.

Answer **Blue scores more; Red is more consistent.**

WORD BANK

median – the typical value

IQR – Q3 − Q1, the spread of the middle half

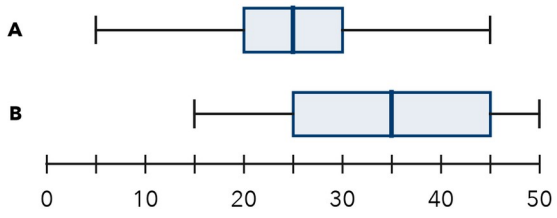
consistent – close together

predictable – you can guess it well

PRACTICE

show your work · label units

1. Reading Minutes. The box plots show minutes read per day by Class A and Class B.



Step 1 Find the line in each box: the medians.

Step 2 IQR = right edge − left edge of each box.

a. Medians: A ___ B ___

b. IQRs: A ___ B ___

c. Usually reads more: _____

d. More consistent: _____

2. Bus Routes. Minutes late on 7 days. **Route 3:** 2, 3, 5, 6, 8, 9, 11. **Route 8:** 0, 1, 4, 6, 10, 14, 15.

Step 1 Order is given. The median is the 4th value.

Step 2 Q1 is the 2nd value, Q3 is the 6th value.

a. Route 3: median ___ IQR ___ Route 8: median ___ IQR ___

b. Which route is more predictable? Why? _____

3. **Plant Heights (cm).** **Sun:** 10, 12, 13, 15, 16, 18, 19, 21. **Shade:** 6, 7, 8, 8, 9, 10, 10, 12.

a. Sun: median ___ cm IQR ___ cm **b.** Shade: median ___ cm IQR ___ cm

4. **Delivery.** Shop K: median 28 min, IQR 8 min. Shop L: median 36 min, IQR 8 min. What is the same?

☐ the same center ☐ the same spread ☐ both center and spread

5. **Find the Error.** Set E: median 70, IQR 25. Set F: median 64, IQR 6. Jada says, "Set E is higher and more consistent." Explain her error.

Frame *Set E has the higher ___. The more consistent set is ___ because its ___ is ___.*

TEST PRACTICE

choose the best answer

6. Group 1: median 12 cm, IQR 3 cm. Group 2: median 12 cm, IQR 9 cm. Which is more consistent?

A Group 1 **B** Group 2 **C** Both the same **D** Cannot tell

7. What is the IQR of 3, 5, 6, 8, 10, 12, 13?

A 7 **B** 8 **C** 10 **D** 17

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home why a bus with a small IQR is easier to plan around.